

2025/2026 exam :

Written (pre-appello and first appello 25/26) :

1. Neurovascular coupling, what is the function of astrocytes in it ?
2. fMRI pre-processing phases
3. Anatomical features and function of the parietal cortex
4. DMN and how do we study brain at rest -
5. Action representation network
6. Phases of memory
7. Pattern of activation, what is it and how can we study it -
8. What Processes that regulate sleep - (+ i biomarker del sonno)
9. describe Ventral and dorsal pathways in vision -
10. How many bodily senses exist?
11. Explain basic PET functioning?
12. What is action potential and how is neural activity supplied with energy?
13. What are neural patterns, what information does it convey, how do we study it?
14. What are the clinical and basic uses of PET?
15. What are Parvocellular and Magnocellular pathways? And what are their functions? -
16. What's the difference between touch and proprioception?
17. What is the use of the prefrontal cortex for behaviour regulation and decision making?
18. DMN what is it? Where is it and how is it related to the external and internal cognition? -
19. What is hebbian plasticity and its role in learning?
20. What's sleep in behavioural and electrophysiology definition. What are the physiological signals to define it?
21. Anatomical features and functions of the temporal lobe.
22. What is the BOLD signal and how is it used for fMRI?

Written (other years):

1. The PET tracers
2. The post processing to be done on the PET image after acquiring the image
3. The fMRI artifacts
4. Which is the role of astrocytes in regulating vascular supply to neurons?
5. Briefly explain the organization of visual pathways
6. What is the so-called "task-positive network" in functional MRI?
7. What is the action recognition network in humans?
8. What is neurovascular coupling?
9. The representation of different semantic categories in the brain?
10. What kind of activity do we find in the transitional stage between wakefulness and sleep?
11. Why are faces relevant?
12. What are the two processes that regulate the level of sleep pressure?
13. The precuneus and the calcarine sulcus
14. What is supramodality?

15. The sulci of medial part of brain
16. Steps to create a scientific study
17. Role of neurotransmitters when we fall asleep
18. Hippocampus and the medical case of its removing
19. Which are the cells involved in the neurovascular coupling?
20. Which are the features of a stimuli that a receptor should inform about?
21. Briefly explain retinotopy of visual areas?
22. How are human faces processed in the brain?
23. What is a "resting state" protocol in functional MRI?
24. What is inf front gyrus, what is there, what are their functions?
25. When are peaks of sleep propensity in circadian rhythm?
26. Homeostatic process, which kind of parameter changes in EEG?
27. Local sleep: which is a condition that happens while the individual is awake, but some regions of the brain go to sleep.
28. Function of Temporal Parietal Junction
29. Effects of prolonged wakefulness
30. How are bodily senses processed in the brain?
31. Describe briefly what are TR and TE
32. Is there a mirror system in the human brain?
33. What is the name of the phenomena consisting of different body parts having different proportions of the cortex?
34. Glymphatic system

Orale(pre-appello) :

- 1.describe the visual pathway
- 2.Describe function of temporal lobe
- 3.What is proprioception, what signals are processed on skin level(pain,pressure,temperature,etc.)
- 4.define flash bulb memory, how encoding happens
- 5.motor descending system
- 6.role of basal ganglia
- 7.sensation's four characteristics and specifically location (density and receptive field)
- 8.factors influencing encoding
9. 4 tactile receptor
10. EEG vs MRI
11. Talk about Thalamus
12. Talk about Hypothalamus
13. Mirror Neuron
- 14: Technique to study Neural pattern (MVPA, pattern correlation Haxby & Pietrini 2001)
15. Action representation network